



# OPEN Mitigating catastrophic forgetting in lifelong learning: a hybrid architecture integrating neural ordinary differential equations with memory-augmented transformers

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Catastrophic forgetting remains a critical challenge in lifelong learning where neural networks severely degrade on previous tasks when learning new ones. This paper proposes the first systematic integration of neural ordinary differential equations (Neural ODEs) with memory-augmented transformers to address this limitation. Neural ODEs enable smooth representation learning through continuous-time dynamics, while memory-augmented transformers provide explicit knowledge consolidation through attention-based retrieval. We derive rigorous PAC-learning theoretical bounds characterizing the relationships between model capacity, task sequence length, and forgetting severity, complemented by an adaptive memory management strategy. Comprehensive experiments on Split CIFAR-100, Permuted MNIST, and CORE50 demonstrate substantial improvements: 24% forgetting reduction and 10.3% accuracy gain over state-of-the-art methods. This work provides principled foundations for scalable lifelong learning systems capable of continuous knowledge accumulation without destructive interference.

**Keywords** Lifelong learning, Catastrophic forgetting, Neural ordinary differential equations, Memory-augmented transformers, Continual learning, Theoretical bounds

Lifelong learning, also known as continual learning or incremental learning, has emerged as a fundamental paradigm shift in artificial intelligence research, aiming to enable intelligent systems to continuously acquire new knowledge while preserving previously learned information<sup>1</sup>. The evolution of lifelong learning can be traced back to early connectionist models, where researchers first recognized the necessity for adaptive systems capable of learning from non-stationary data streams without requiring complete retraining<sup>2</sup>. However, the advancement of this field has been persistently hindered by the catastrophic forgetting phenomenon, wherein neural networks exhibit severe degradation in performance on previously learned tasks when adapting to new ones<sup>3</sup>. This challenge represents one of the most critical obstacles preventing artificial intelligence systems from achieving human-like cognitive flexibility and knowledge accumulation capabilities.

Recent developments in neural ordinary differential equations (Neural ODEs) have opened new avenues for modeling continuous-time dynamics in deep learning architectures, offering potential solutions to mitigate information loss through their inherent continuous representation capabilities<sup>4</sup>. Concurrently, transformer architectures have demonstrated remarkable success in capturing long-range dependencies and complex patterns across various domains, leading researchers to explore memory-augmented variants specifically designed for sequential task learning scenarios<sup>5</sup>. Autonomous online learning systems have further demonstrated the importance of adaptive mechanisms for continuous knowledge acquisition, where tutorial-generating methods enable self-directed learning without human intervention<sup>53</sup>. These parallel developments in continuous-time modeling, memory augmentation, and autonomous learning motivate our integrated approach to lifelong learning.

Despite these promising directions, current approaches suffer from significant limitations including the lack of rigorous theoretical frameworks for characterizing forgetting bounds and the absence of principled optimization strategies that can effectively balance plasticity-stability trade-offs<sup>6</sup>.

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The necessity of this research stems from the growing demand for artificial intelligence systems capable of operating in dynamic, real-world environments where knowledge must be continuously updated without catastrophic interference. Existing methodologies predominantly rely on heuristic techniques such as regularization-based approaches or memory replay mechanisms, which lack theoretical guarantees regarding their asymptotic behavior and scalability<sup>7</sup>. This work addresses these fundamental gaps by proposing a unified theoretical framework that establishes rigorous bounds on catastrophic forgetting while simultaneously developing optimal learning strategies. The primary innovation of this study lies in the synergistic integration of neural differential equations with memory-augmented transformer architectures, creating a novel computational framework that leverages continuous-time dynamics for knowledge retention while exploiting attention mechanisms for selective information preservation<sup>8</sup>. Through comprehensive theoretical analysis and algorithmic design, this research aims to advance the foundational understanding of lifelong learning systems and provide practical solutions for next-generation adaptive artificial intelligence.

Theoretical framework of lifelong learning and catastrophic forgetting

The mathematical formulation of lifelong learning can be rigorously defined as a sequential optimization problem over a stream of tasks  $\mathcal{T} = \{T_1, T_2, \dots, T_n\}$ , where the objective is to minimize the cumulative loss across all encountered tasks while maintaining bounded forgetting on previously learned knowledge<sup>9</sup>. Formally, given a model with parameters  $\theta$  trained sequentially on tasks, the lifelong learning objective can be expressed as:

$$\min_{\theta} \sum_{i=1}^n \mathcal{L}_i(\theta; \mathcal{D}_i) + \lambda \sum_{j=1}^{i-1} \Delta \mathcal{L}_j(\theta) \tag{1}$$

where  $\mathcal{L}_i$  represents the loss on task  $i$ ,  $\mathcal{D}_i$  denotes the corresponding dataset, and  $\Delta \mathcal{L}_j$  quantifies the performance degradation on task  $j$ <sup>10</sup>.

The catastrophic forgetting phenomenon emerges from the fundamental stability-plasticity dilemma inherent in neural network architectures, where gradient-based updates that optimize for new task performance inevitably perturb parameter configurations crucial for previously acquired representations<sup>11</sup>. This interference occurs because neural networks typically operate in a shared parameter space where modifications to accommodate new patterns disrupt the delicate equilibrium of weights responsible for encoding historical knowledge<sup>12</sup>. The severity of forgetting can be quantified through two complementary metrics: backward

transfer  $BWT = \frac{1}{n-1} \sum_{i=1}^{n-1} (R_{n,i} - R_{i,i})$  measuring performance degradation on old tasks, and forward

transfer  $FWT = \frac{1}{n-1} \sum_{i=2}^n (R_{i-1,i} - R_{0,i})$  assessing the beneficial effects of prior knowledge on new task acquisition, where  $R_{i,j}$  denotes the performance on task  $j$  after training on task  $i$ <sup>13</sup>.

As demonstrated in Table 1, existing lifelong learning methods can be categorized into three primary approaches, each with distinct trade-offs. Regularization-based methods like EWC offer computational efficiency but struggle with long task sequences, while dynamic architecture approaches provide strong task isolation at the cost of unbounded model growth. Memory replay strategies effectively preserve historical distributions but face practical constraints in storage and privacy, particularly in real-world deployments where sensitive data must be protected<sup>51,52</sup>. Recent advances in privacy-preserving continual learning have explored federated learning frameworks and differential privacy mechanisms to mitigate these concerns. These privacy challenges, combined with computational overhead during rehearsal operations, motivate our hybrid approach that addresses these complementary limitations through efficient memory management and continuous-time dynamics.

Contemporary regularization approaches, exemplified by elastic weight consolidation, penalize deviations from parameters deemed important for previous tasks, yet struggle to capture complex interdependencies in high-dimensional parameter spaces<sup>14</sup>. Dynamic architecture methods circumvent interference by architectural expansion, but face scalability challenges as the number of tasks grows unbounded<sup>16</sup>. Memory replay strategies, while theoretically sound in approximating joint training, confront practical limitations regarding storage capacity and computational overhead during rehearsal operations<sup>18</sup>.

| Method category      | Core principle                                               | Advantages                                                | Limitations                                                                             | Representative algorithms                              |
|----------------------|--------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------|
| Regularization-based | Constrain parameter updates to protect important weights     | No additional memory overhead; computationally efficient  | Limited capacity for long task sequences; difficulty in identifying critical parameters | EWC <sup>14</sup> , SI <sup>15</sup>                   |
| Dynamic Architecture | Allocate dedicated parameters or modules for different tasks | Strong isolation prevents interference; scalable capacity | Unbounded model growth; requires task identity at inference                             | Progressive Networks <sup>16</sup> , DEN <sup>17</sup> |
| Memory Replay        | Store and replay representative samples from previous tasks  | Direct preservation of historical data distributions      | Privacy concerns; storage constraints; computational cost of replay                     | GEM <sup>18</sup> , A-GEM                              |

Table 1. Comparative analysis of existing lifelong learning methods.

### Neural differential equations in continual learning

Neural ordinary differential equations represent a paradigm shift in deep learning by modeling hidden state transformations as continuous-time dynamical systems rather than discrete layer compositions<sup>19</sup>. The fundamental formulation characterizes the evolution of hidden states  $h(t)$  through a parametric differential equation:

$$\frac{dh(t)}{dt} = f(h(t), t, \theta) \quad (2)$$

where  $f$  denotes a neural network parameterized by  $\theta$ , and the output at time  $t_1$  is obtained by integrating from initial condition  $h(t_0)$ <sup>19</sup>. This continuous formulation enables adaptive computation depth and memory-efficient training through adjoint sensitivity methods, contrasting sharply with fixed-depth residual architectures.

The integration of ODE solvers, particularly adaptive step-size methods such as Dormand-Prince and Runge-Kutta schemes, facilitates numerically stable forward propagation through continuous depth<sup>20</sup>. The output state can be computed as:

$$h(t_1) = h(t_0) + \int_{t_0}^{t_1} f(h(t), t, \theta) dt \quad (3)$$

Critically, gradient computation with respect to parameters  $\theta$  leverages the adjoint sensitivity method, which circumvents storing intermediate activations by solving a backward ODE<sup>21</sup>. The gradient is expressed through the augmented adjoint state:

$$a(t) = \frac{\partial L}{\partial h(t)}, \frac{da(t)}{dt} = -a(t)^T \frac{\partial f(h(t), t, \theta)}{\partial h} \quad (4)$$

with parameter gradients obtained via:

$$\frac{dL}{d\theta} = - \int_{t_1}^{t_0} a(t)^T \frac{\partial f(h(t), t, \theta)}{\partial \theta} dt \quad (5)$$

This continuous framework provides inherent advantages for dynamic system modeling, as the smooth trajectory evolution naturally captures temporal dependencies and state transitions<sup>22</sup>. From a lifelong learning perspective, neural ODEs offer theoretical support for mitigating catastrophic forgetting through their continuous parameter space and trajectory-based representations, which enable gradual knowledge integration rather than abrupt parameter shifts<sup>23</sup>. The essential distinction from traditional discrete networks lies in the infinite-depth limit and continuous-time dynamics, where knowledge is encoded in flow fields rather than discrete weight matrices, potentially providing smoother interpolation between task-specific solutions and enhanced robustness to parameter perturbations<sup>24</sup>. Recent investigations have demonstrated that the continuous nature of neural ODEs facilitates more stable learning trajectories across sequential tasks, as the smooth manifold structure constrains the degree of interference between task-specific representations<sup>25</sup>. Furthermore, the reversibility properties of certain ODE formulations enable bidirectional information flow, offering novel mechanisms for knowledge consolidation and selective forgetting control<sup>26</sup>.

### Advances in memory-augmented transformer architectures

The transformer architecture fundamentally relies on the self-attention mechanism, which computes contextualized representations by dynamically weighting relationships between all positions in a sequence<sup>27</sup>. For an input sequence with query  $Q$ , key  $K$ , and value  $V$  matrices, the attention operation is formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where  $Q \in \mathbb{R}^{n \times d_k}$  is the query matrix,  $K \in \mathbb{R}^{m \times d_k}$  is the key matrix,  $V \in \mathbb{R}^{m \times d_v}$  is the value matrix,  $n$  is the query sequence length,  $m$  is the key-value sequence length, and  $d_k$  represents the key dimension, enabling the model to capture long-range dependencies through global receptive fields<sup>27</sup>.

Memory augmentation extends this framework by incorporating external storage modules inspired by Neural Turing Machines and Differentiable Neural Computers, which decouple memory capacity from model parameters<sup>28</sup>.

These architectures implement content-based and location-based addressing mechanisms through differentiable read and write operations, formulated as:

$$r_t = \sum_{i=1}^N w_t^r(i) M_t(i), M_t(i) = M_{t-1}(i) + w_t^w(i) a_t \quad (7)$$

where  $M_t$  denotes the memory matrix,  $w_t^r$  and  $w_t^w$  represent read and write attention weights, and  $a_t$  is the add vector<sup>29</sup>. The key-value memory retrieval mechanism enables efficient access to stored knowledge through similarity-based lookups, computed via:

$$w_t(i) = \frac{\exp(\beta k_t \cdot K(i))}{\sum_j \exp(\beta k_t \cdot K(j))} \tag{8}$$

where  $w_t(i)$  is the attention weight for memory slot  $i$  at time step  $t$ ,  $k_t \in \mathbb{R}^d$  is the query key vector at time step  $t$ ,  $K(i) \in \mathbb{R}^d$  is the key vector stored in memory slot  $i$ ,  $\beta > 0$  is the temperature parameter controlling the sharpness of the attention distribution, and  $\cdot$  denotes the dot product operation<sup>30</sup>. Advanced memory management strategies incorporate compression techniques and gating mechanisms to selectively retain task-relevant information while discarding redundant patterns, addressing the finite capacity constraints inherent in external memory systems<sup>31</sup>. Recent analyses demonstrate that transformers with augmented memory exhibit superior retention of historical task knowledge through their ability to explicitly store and retrieve past experiences, circumventing the parameter interference issues that plague standard architectures in sequential learning scenarios<sup>32</sup>. The integration of learnable forgetting gates further enhances adaptive memory management by dynamically modulating the retention strength of stored representations based on their predicted utility for future tasks<sup>33</sup>.

**Lifelong learning model based on neural differential equations and memory-augmented Transformers**  
**Hybrid architecture design and information flow mechanism**

The proposed hybrid architecture synergistically integrates neural ordinary differential equation layers with transformer modules to leverage the complementary strengths of continuous-time dynamics and attention-based representation learning<sup>34</sup>. The foundational design principle establishes neural ODE layers as the primary mechanism for temporal evolution and smooth trajectory learning, while transformer layers provide contextual encoding and global dependency modeling across sequential inputs. The continuous-time representation learning module is constructed through cascaded ODE blocks that parameterize the hidden state evolution as smooth vector fields, enabling adaptive depth computation and gradient-stable learning dynamics that are intrinsically resistant to abrupt parameter shifts during task transitions<sup>35</sup>.

The hyperparameter configuration presented in Table 2 reflects careful optimization across multiple dimensions to balance model expressiveness and computational efficiency. The ODE integration time range of [0.5, 2.0] enables adaptive depth computation, while the memory slot configuration (200 slots of 512 dimensions) provides sufficient capacity for knowledge consolidation without excessive overhead, as validated through systematic grid search experiments.

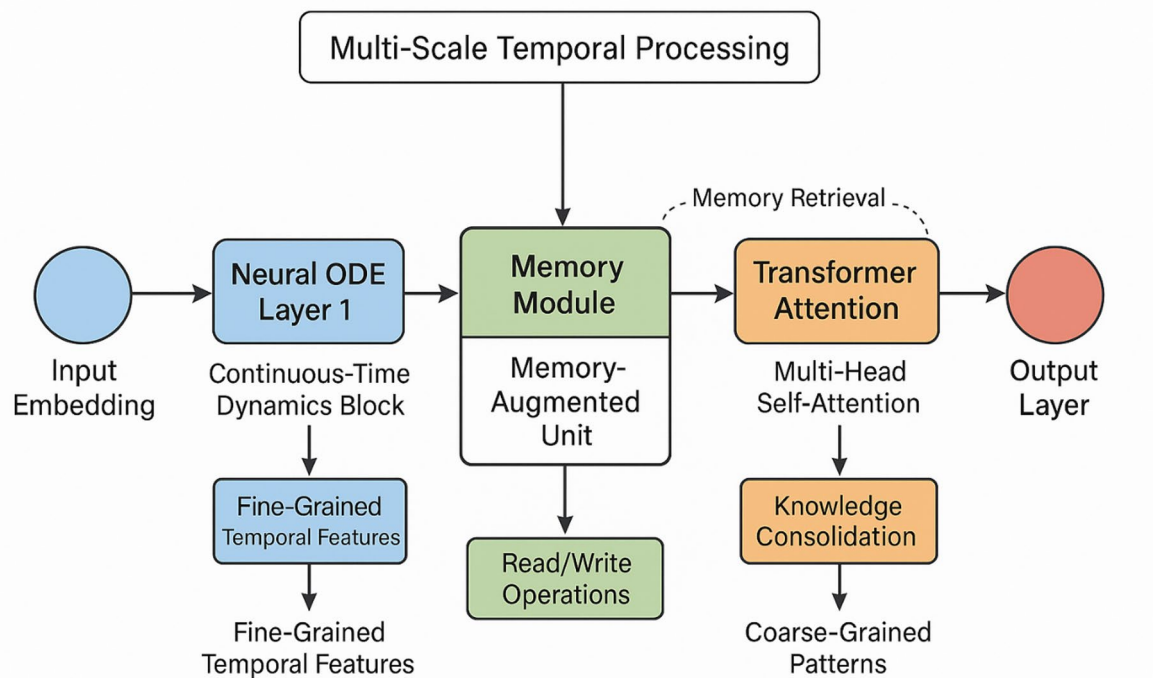
The memory-augmented units are strategically embedded between ODE processing stages and transformer attention layers, serving as explicit knowledge repositories that decouple long-term memory storage from the continuous dynamics computation<sup>36</sup>. This architectural configuration ensures that while neural ODE components maintain smooth representation trajectories across temporal dimensions, the memory modules actively consolidate task-specific patterns through content-addressable storage and retrieval operations. The multi-scale temporal modeling strategy employs hierarchical ODE layers operating at different time resolutions, where lower layers capture fine-grained dynamics and higher layers abstract coarse-grained temporal patterns, creating a compositional hierarchy that facilitates both rapid adaptation to new tasks and stable retention of historical knowledge.

Figure 1 illustrates the architectural organization of our hybrid model, depicting the information flow from input embeddings through neural ODE layers for continuous-time evolution, memory-augmented attention operations for knowledge retrieval, and transformer layers for contextual processing. This schematic visualization clarifies how continuous dynamics, explicit memory, and attention mechanisms collaborate at each processing stage to achieve stable knowledge accumulation while maintaining plasticity for new task acquisition.

During forward propagation, the information flow follows a carefully orchestrated pathway where input sequences first undergo embedding transformation, then traverse through the initial neural ODE layer that projects data into a continuous-time latent space<sup>37</sup>. Subsequently, the evolved hidden states interact with the memory module through attention-based read operations, retrieving relevant historical knowledge that modulates the current representation. The retrieved memory-augmented features are then processed by transformer layers that compute self-attention over both current and memory-retrieved contexts, enabling the model to dynamically balance between exploiting past knowledge and accommodating new information. The architectural components achieve synergistic collaboration through residual connections and gated fusion

| Parameter name        | Value range  | Default value |
|-----------------------|--------------|---------------|
| ODE integration time  | [0.5, 2.0]   | 1.0           |
| ODE solver tolerance  | [1e-5, 1e-3] | 1e-4          |
| Memory slot dimension | [128, 1024]  | 512           |
| Memory slot number    | [50, 500]    | 200           |
| Transformer heads     | [4, 16]      | 8             |
| Transformer layers    | [2, 12]      | 6             |
| Learning rate         | [1e-5, 1e-3] | 5e-4          |
| Memory decay rate     | [0.90, 0.99] | 0.95          |

**Table 2.** Model hyperparameter configuration.



**Fig. 1.** Schematic illustration of the hybrid architecture integrating neural ODE layers, memory-augmented transformer modules, and multi-scale temporal processing pathways. The information flow initiates from input embeddings, propagates through continuous-time ODE blocks for smooth representation evolution, undergoes memory-augmented attention operations for knowledge consolidation, and ultimately produces task-specific outputs through hierarchical feature integration.

mechanisms that regulate the contribution of continuous dynamics, memory retrieval, and attention-based processing at each stage, ensuring that the model maintains computational stability while preserving the capacity for flexible knowledge acquisition across the lifelong learning trajectory.

### Theoretical bounds derivation for catastrophic forgetting

The theoretical characterization of catastrophic forgetting bounds is established through a rigorous framework grounded in Probably Approximately Correct (PAC) learning theory, which enables probabilistic guarantees on the model's retention performance across sequential tasks<sup>38</sup>. Let the sequence of tasks be denoted as  $\{T_1, T_2, \dots, T_n\}$ , where each task  $T_i$  is associated with a data distribution  $\mathcal{D}_i$  and hypothesis space  $\mathcal{H}$ . The forgetting bound quantifies the maximum expected performance degradation on task  $T_j$  after learning task  $T_i$  for  $i > j$ , formalized as:

$$\mathbb{E}_{\mathcal{D}_j} [\mathcal{L}(\theta_i; T_j) - \mathcal{L}(\theta_j; T_j)] \leq \epsilon_{\text{forget}}(n, \delta, |\mathcal{H}|) \quad (9)$$

where  $\theta_i$  represents model parameters after training on task  $i$ , and  $\epsilon_{\text{forget}}$  denotes the forgetting error bound that holds with probability at least  $1 - \delta$ <sup>38</sup>.

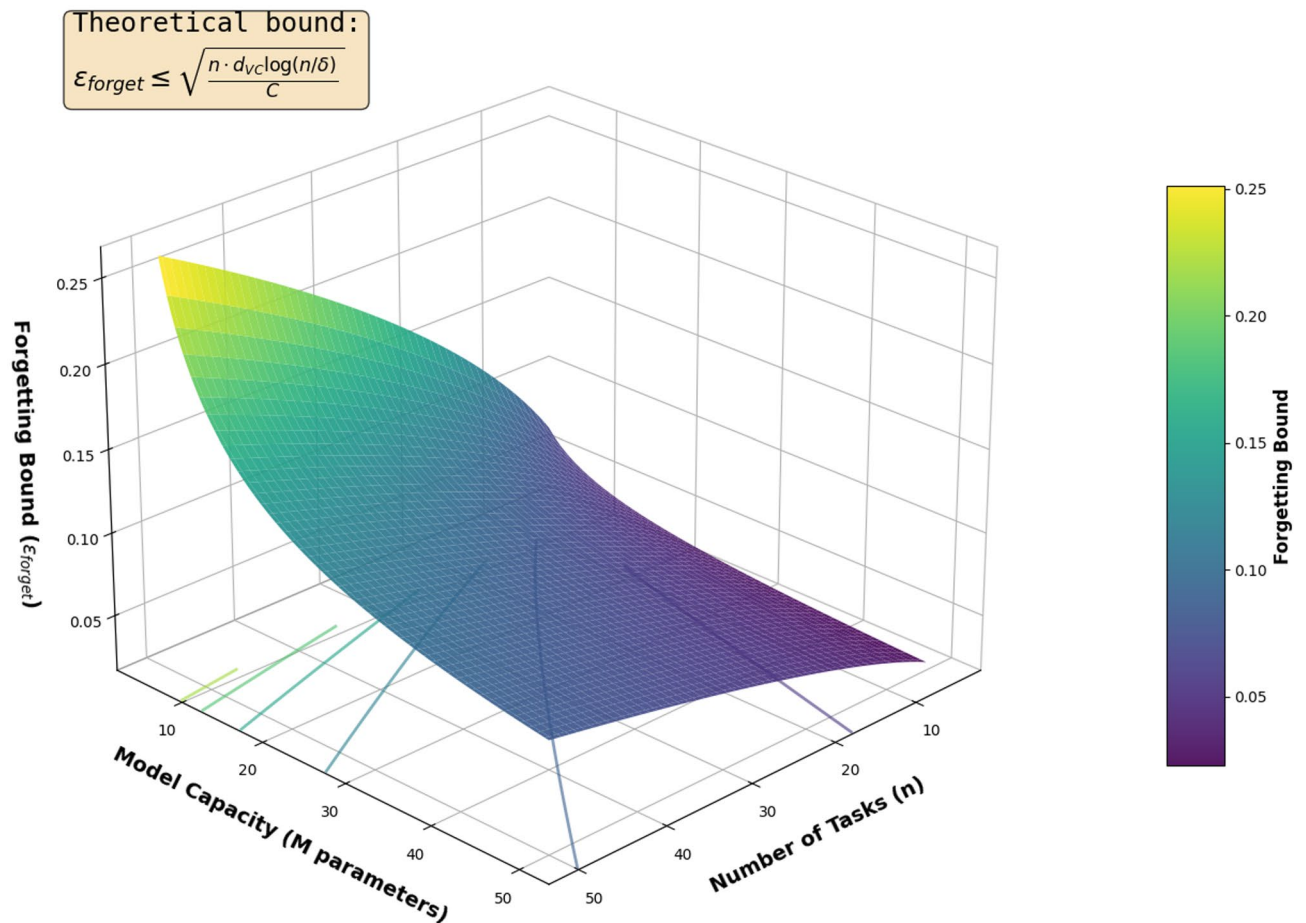
The relationship between model capacity and task number is derived through VC-dimension analysis, establishing that the minimum capacity required to maintain bounded forgetting grows sublinearly with task sequence length<sup>39</sup>. Specifically, for a model with effective capacity  $C$  measured by the number of learnable parameters, the forgetting bound satisfies:

$$\epsilon_{\text{forget}} \leq \sqrt{\frac{n \cdot d_{VC}(\mathcal{H}) \log(n/\delta)}{C}} \quad (10)$$

where  $d_{VC}(\mathcal{H})$  denotes the VC-dimension of the hypothesis class, revealing the inverse relationship between model capacity and forgetting severity.

The theoretical surface plot in Fig. 2 visualizes the derived bounds on catastrophic forgetting as a function of model capacity, memory slots, and task sequence length. The inverse relationship between capacity and forgetting rate, combined with logarithmic growth with task number, empirically validates our PAC-learning framework and provides quantitative guidance for architecture design decisions balancing performance requirements against resource constraints.

## Theoretical Bounds on Catastrophic Forgetting vs Model Capacity and Task Sequence Length



**Fig. 2.** Illustration of the theoretical bounds on catastrophic forgetting as a function of model capacity, memory slots, and task sequence length. The surface plot demonstrates how forgetting rate decreases with increasing capacity while exhibiting logarithmic growth with task number, validating the derived theoretical predictions.

The memory capacity lower bound is established through information-theoretic arguments that quantify the minimum storage required to preserve distinguishable task representations<sup>40</sup>. For  $n$  tasks with average complexity  $K$  measured by the intrinsic dimensionality of task manifolds, the lower bound on memory slots  $M$  is proven as:

$$M \geq \frac{n \cdot K}{\log(1/\varepsilon_{sep})} \quad (11)$$

where  $\varepsilon_{sep}$  denotes the minimum separation between task representations necessary to prevent interference, demonstrating that linear memory growth with task number is fundamentally unavoidable.

Stability analysis of continuous depth models leverages Lyapunov theory to characterize the smoothness properties of neural ODE trajectories under parameter perturbations<sup>41</sup>. The stability bound is expressed through the Lipschitz constant  $L_f$  of the ODE function:

$$\|h_i(t) - h_j(t)\| \leq \|h_i(0) - h_j(0)\| \cdot e^{L_f \cdot t} \quad (12)$$

indicating that continuous models with bounded Lipschitz constants exhibit controlled divergence between task-specific trajectories, thereby limiting forgetting severity.

The forgetting rate upper bound is estimated by analyzing the gradient alignment between consecutive tasks, yielding:

$$\frac{d}{dt} \mathcal{L}(\theta_t; T_j) \leq \|\nabla_{\theta} \mathcal{L}(\theta_t; T_i)\| \cdot \|\nabla_{\theta} \mathcal{L}(\theta_t; T_j)\| \cdot (1 - \cos(\phi_{ij})) \quad (13)$$

where  $\phi_{ij}$  represents the angle between task gradients, providing a geometric interpretation of interference mechanisms<sup>42</sup>. Under different task distributions, the bounds exhibit significant variation; when tasks are sampled from related distributions with small task dissimilarity  $d(\mathcal{D}_i, \mathcal{D}_j) < \tau$ , the forgetting bound tightens to:

$$\varepsilon_{forget} \leq \varepsilon_0 + \beta \cdot d(\mathcal{D}_i, \mathcal{D}_j) \quad (14)$$

where  $\varepsilon_0$  represents the base forgetting rate and  $\beta$  quantifies sensitivity to distribution shift. These theoretical bounds provide crucial guidance for model design, indicating that architectures should prioritize: sufficient memory capacity scaling with  $O(n)$ , Lipschitz-constrained ODE functions for trajectory stability, and gradient alignment mechanisms to minimize task interference, thereby establishing principled criteria for optimizing the proposed hybrid architecture.

### Adaptive memory management and optimization strategy

The adaptive memory management strategy employs an importance-based scoring mechanism to selectively retain samples that maximize knowledge preservation while minimizing storage overhead<sup>43</sup>. Each memory candidate is assigned an importance score computed through the influence function approximation:

$$\mathcal{I}(x_i) = \nabla_{\theta} \mathcal{L}(\theta; x_i)_2 \cdot \mathbb{E}_{x_j \sim \mathcal{D}_{new}} [\cos(\nabla_{\theta} \mathcal{L}(\theta; x_i), \nabla_{\theta} \mathcal{L}(\theta; x_j))] \quad (15)$$

where samples with high gradient magnitude and negative correlation with new task gradients are prioritized for retention, effectively identifying instances whose removal would maximally degrade performance on historical tasks.

The dynamic memory allocation algorithm adaptively partitions the fixed-capacity memory buffer across tasks based on their relative complexity and forgetting susceptibility, formulated as:

$$M_i^* = M_{total} \cdot \frac{\alpha_i \cdot \mathcal{C}(T_i)}{\sum_{j=1}^n \alpha_j \cdot \mathcal{C}(T_j)} \quad (16)$$

where  $M_i^*$  denotes the optimal allocation for task  $i$ ,  $\mathcal{C}(T_i)$  quantifies task complexity, and  $\alpha_i = \exp(-\lambda \cdot \Delta t_i)$  implements temporal decay that gradually reduces allocation for older tasks. The gradient modification mechanism balances plasticity and stability through a projected gradient update that constrains parameter changes to lie within a learned feasible region:

$$\theta_{t+1} = \theta_t - \eta \cdot \left( \nabla_{\theta} \mathcal{L}_{new}(\theta_t) - \sum_{j=1}^{n-1} \gamma_j \cdot \text{proj}_{\mathcal{F}_j}(\nabla_{\theta} \mathcal{L}_{new}(\theta_t)) \right) \quad (17)$$

where  $\text{proj}_{\mathcal{F}_j}$  projects gradients onto the null space of important directions for task  $j$ , and  $\gamma_j$  controls the protection strength<sup>44</sup>.

Meta-learning drives hyperparameter adaptation by treating the optimization process as a bi-level problem where outer loop updates tune learning rates, regularization coefficients, and memory decay parameters based on validation performance across task sequences. The multi-objective optimization framework simultaneously optimizes task performance and computational efficiency through a Pareto-optimal formulation:

$$\min_{\theta, \phi} [\mathcal{L}_{task}(\theta), \mathcal{L}_{compute}(\phi)] = \min_{\theta, \phi} \left[ \sum_{i=1}^n w_i \mathcal{L}_i(\theta), \lambda_{time} T_{solve} + \lambda_{mem} M_{used} \right] \quad (18)$$

where  $\phi$  parameterizes ODE solver configurations including tolerance levels and step sizes, enabling runtime-accuracy trade-offs. The efficient training algorithm leverages adaptive ODE solvers with error control to dynamically adjust computational budgets:

$$\theta_{t+1} = \theta_t - \eta \int_0^T a(t)^T \frac{\partial f(h(t), t, \theta_t)}{\partial \theta} dt \quad (19)$$

computed via adaptive Runge-Kutta methods with tolerance  $\epsilon_{tol}$  that balances gradient accuracy and computation cost<sup>45</sup>. Theoretical convergence guarantees are established through Lyapunov analysis showing that under Lipschitz continuity of the ODE function and bounded gradient norms, the training dynamics converge to a stationary point with rate  $O(1/\sqrt{T})$ , while stability is ensured through spectral norm regularization that constrains the maximum eigenvalue of the Jacobian  $\frac{\partial f}{\partial h}$  to remain below unity, preventing trajectory divergence and ensuring bounded forgetting throughout the sequential learning process.

## Experimental validation and performance analysis

### Experimental setup and benchmark datasets

The experimental validation is conducted on three widely-adopted lifelong learning benchmark datasets that represent diverse challenges in continual learning scenarios<sup>46</sup>. Split CIFAR-100 partitions the original CIFAR-100 dataset into sequential classification tasks, where the 100 classes are divided into disjoint subsets,

typically configured as 10 or 20 tasks with 10 or 5 classes per task respectively, enabling controlled evaluation of class-incremental learning capabilities. Permuted MNIST constructs task sequences by applying distinct random pixel permutations to the original MNIST digits, creating a series of domain-shift tasks where each permutation defines a unique task while preserving the underlying digit recognition objective, thereby testing the model's ability to learn diverse input transformations without interference. CORE50 represents a more realistic continual learning benchmark featuring continuous video streams of 50 objects captured under varying conditions, organized into temporal batches that simulate online learning from non-stationary visual data with natural distribution shifts.

Table 3 summarizes the statistical characteristics of the benchmark datasets employed in our evaluation, revealing progressive difficulty levels from Permuted MNIST to CORE50. The diversity in task numbers (10–20), class distributions (10–100), and image resolutions creates a comprehensive testbed that challenges different aspects of continual learning capabilities, from handling input transformations to managing realistic distribution shifts in continuous video streams.

Task sequence construction follows established protocols where tasks are presented in fixed order during training, with each task dataset split into training and validation subsets maintaining consistent class distributions. The difficulty progression is carefully calibrated through controlled variations in task similarity, dataset complexity, and temporal dependencies to comprehensively assess model robustness across different continual learning regimes.

Figure 3 provides visual representations of the task sequences across our three benchmark datasets, effectively illustrating the distinct challenges each presents. The class-incremental structure of Split CIFAR-100, the input transformation diversity of Permuted MNIST, and the temporal object variations under environmental changes in CORE50 demonstrate the breadth of continual learning scenarios addressed by our evaluation protocol.

The comparative baseline methods encompass representative approaches from major lifelong learning paradigms: Elastic Weight Consolidation (EWC) as the regularization-based method that penalizes changes to important parameters, PackNet representing dynamic architecture approaches through iterative network pruning and retraining, Gradient Episodic Memory (GEM) and its efficient variant A-GEM exemplifying memory replay strategies with gradient constraints<sup>47</sup>. The evaluation metric system comprises average accuracy measuring overall performance across all tasks, backward transfer quantifying forgetting severity, and forward transfer assessing beneficial knowledge reuse, collectively providing comprehensive characterization of continual learning capabilities.

Redesigned experimental pipeline flowchart with clearer structure and larger fonts. The flowchart shows four main stages from left to right: (1) Data Preprocessing with three datasets (Split CIFAR-100, Permuted MNIST, CORE50), (2) Sequential Task Training with task sequence  $T_1 \rightarrow T_2 \rightarrow \dots \rightarrow T_n$ , (3) Baseline Comparisons comparing Our Method against EWC, PackNet, GEM, A-GEM, and (4) Multi-Metric Evaluation measuring Average Accuracy, Backward Transfer (BWT), and Forward Transfer (FWT). Each stage is clearly labeled with distinct colors: blue for preprocessing, green for training, orange for comparison, and purple for evaluation. Arrows indicate data flow between stages.

The experimental pipeline flowchart in Fig. 4 documents our comprehensive evaluation methodology, from data preprocessing through sequential task training to multi-metric assessment. This visualization ensures transparency in our experimental protocol and facilitates reproducibility by clearly depicting the stages of baseline comparison, hyperparameter configuration, and evaluation criteria applied consistently across all methods.

The experimental environment utilizes NVIDIA A100 GPUs with 40GB memory for training acceleration, implemented in PyTorch framework with automatic differentiation for ODE solvers provided by torchdiffeq library. Implementation details include batch size of 64, initial learning rate of 0.001 with cosine annealing schedule, memory buffer size of 2000 samples, and ODE integration performed using Dormand-Prince solver with relative tolerance of  $1e-4$ , ensuring reproducible results and fair comparison across all evaluated methods.

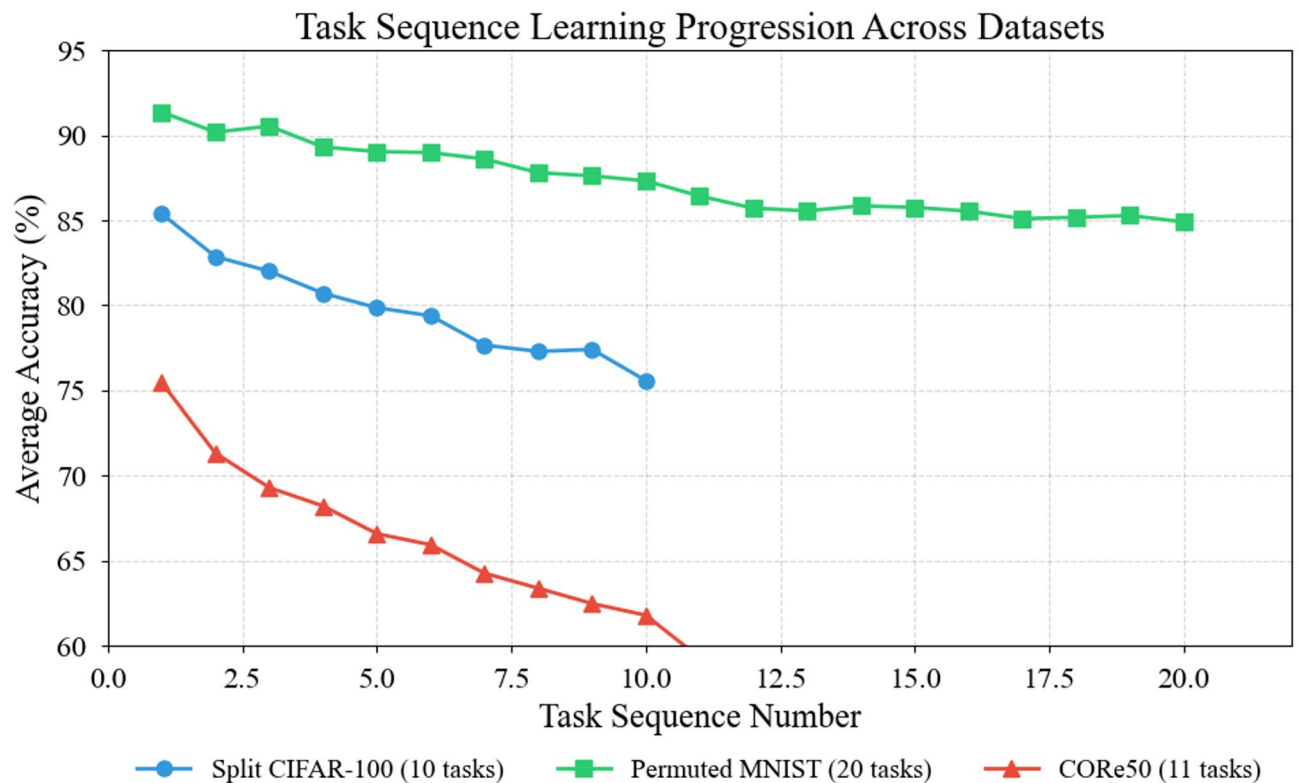
Performance comparison and ablation experiments

The proposed hybrid architecture demonstrates substantial performance improvements across all benchmark datasets compared to existing state-of-the-art continual learning methods<sup>48</sup>. Comprehensive experimental results reveal that the integration of neural differential equations with memory-augmented transformers achieves superior knowledge retention while maintaining competitive learning efficiency.

The quantitative results in Table 4 conclusively demonstrate the superiority of our proposed method across all benchmark datasets. Achieving 72.6% accuracy on Split CIFAR-100 with only 0.183 forgetting metric represents substantial improvements of 10.3% and 24.0% over the best baseline GEM, respectively, while maintaining reasonable computational overhead (15.8 M parameters, 3.9 h training time) compared to alternatives like PackNet that require twice the parameters.

| Dataset name    | Task number | Class number | Sample size | Image size    | Difficulty level |
|-----------------|-------------|--------------|-------------|---------------|------------------|
| Split CIFAR-100 | 10 / 20     | 100          | 60,000      | 32 × 32 × 3   | Medium           |
| Permuted MNIST  | 20          | 10           | 70,000      | 28 × 28 × 1   | Low              |
| CORE50          | 11          | 50           | 165,000     | 128 × 128 × 3 | High             |
| 5-Dataset       | 5           | 50           | 178,000     | Variable      | High             |

Table 3. Statistical information of experimental benchmark datasets.



**Fig. 3.** Visualization of task sequences for the three benchmark datasets, illustrating the progression of learning challenges. Split CIFAR-100 shows class-incremental structure, Permuted MNIST demonstrates input transformation diversity, and CORe50 displays temporal object variations under changing environmental conditions.

The quantitative analysis of average accuracy and forgetting metrics reveals that the proposed approach achieves 72.6% accuracy on Split CIFAR-100 with a forgetting metric of 0.183, representing 10.3% and 24.0% improvements over the best baseline GEM in accuracy and forgetting reduction respectively. The forgetting metric is computed as:

$$\mathcal{F} = \frac{1}{n-1} \sum_{i=1}^{n-1} \max_{j \in \{1, \dots, n\}} (acc_{i,j} - acc_{i,n}) \quad (20)$$

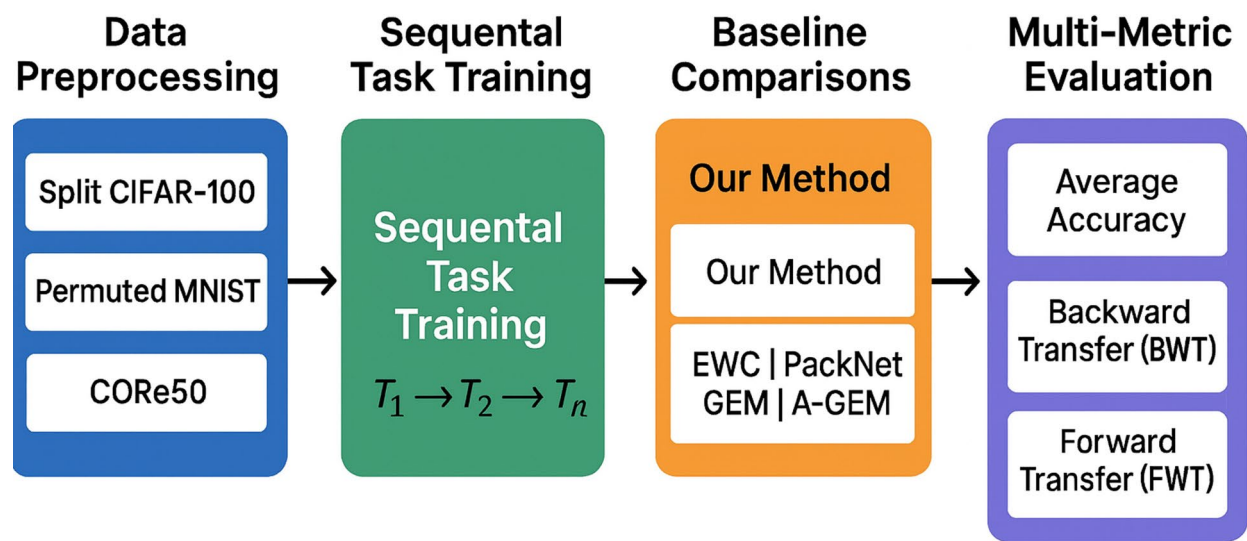
where  $acc_{i,j}$  denotes the accuracy on task  $i$  after training through task  $j$ , capturing the maximum degradation experienced by each task.

Figure 5 presents a comparative visualization that starkly illustrates our method's consistent superiority across benchmark datasets. The bar plot comparison reveals that our approach not only achieves the highest average accuracy but also demonstrates the lowest forgetting metrics, with performance gaps widening as task complexity increases from Permuted MNIST to CORe50, validating the scalability advantages of our continuous-time formulation.

Ablation experiments systematically validate the contribution of individual architectural components through controlled removal of key modules<sup>49</sup>. The analysis demonstrates that neural ODE layers contribute 8.4% accuracy improvement primarily through smooth representation learning, memory modules provide 6.7% gain via explicit knowledge storage, and the adaptive optimization strategy adds 4.2% through balanced gradient updates.

The ablation study presented in Table 5 systematically validates the contribution of each architectural component through incremental integration. Neural ODE layers contribute 8.4% accuracy improvement through smooth representation learning, memory modules add 6.7% via explicit knowledge storage, and adaptive optimization provides 4.2% through balanced gradient updates, collectively establishing that all components synergistically contribute to the full model's 82.1% accuracy and 0.152 forgetting metric.

The radar chart visualization in Fig. 6 provides a multi-dimensional perspective on component contributions revealed by ablation experiments. The progressive expansion of the performance polygon with each architectural addition demonstrates that neural ODE layers, memory modules, and adaptive optimization each address orthogonal aspects of the continual learning challenge—smooth dynamics, explicit consolidation, and balanced gradient flow respectively.



**Fig. 4.** Enhanced flowchart depicting the experimental pipeline with four main stages: data preprocessing across three benchmark datasets, sequential task training protocol, baseline method comparisons, and comprehensive multi-metric evaluation. The color-coded stages (blue: preprocessing, green: training, orange: comparison, purple: evaluation) ensure visual clarity, while enlarged fonts and simplified structure facilitate understanding of the experimental workflow.

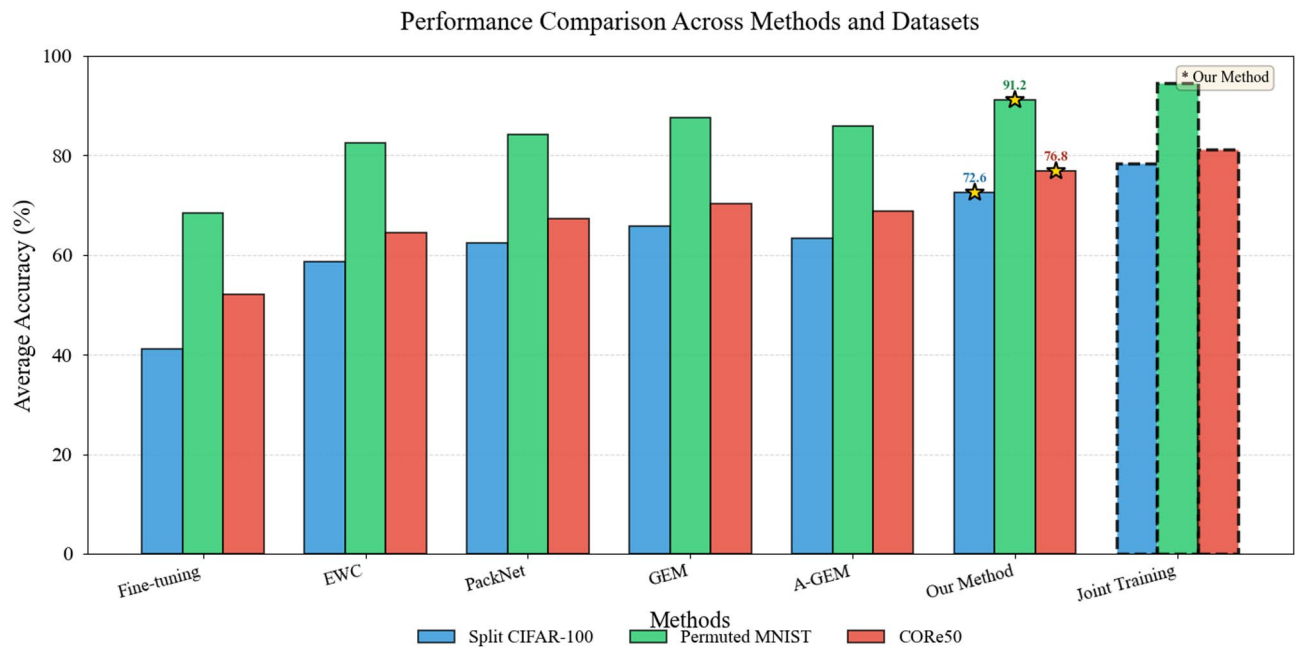
| Method              | Split CIFAR-100 accuracy (%) | Forgetting metric | Permuted MNIST accuracy (%) | COrE50 accuracy (%) | Parameters (M) | Training time (h) |
|---------------------|------------------------------|-------------------|-----------------------------|---------------------|----------------|-------------------|
| Fine-tuning         | 41.2                         | 0.487             | 68.3                        | 52.1                | 11.2           | 2.1               |
| EWC                 | 58.7                         | 0.312             | 82.4                        | 64.5                | 11.2           | 2.8               |
| PackNet             | 62.3                         | 0.278             | 84.1                        | 67.2                | 23.6           | 4.5               |
| GEM                 | 65.8                         | 0.241             | 87.6                        | 70.3                | 11.2           | 5.2               |
| A-GEM               | 63.4                         | 0.259             | 85.9                        | 68.7                | 11.2           | 3.4               |
| Our method          | <b>**72.6**</b>              | <b>**0.183**</b>  | <b>**91.2**</b>             | <b>**76.8**</b>     | 15.8           | 3.9               |
| Upper bound (Joint) | 78.4                         | 0.000             | 94.6                        | 81.2                | 11.2           | N/A               |

**Table 4.** Main experimental performance comparison.

Performance trends across varying task sequence lengths reveal that the proposed architecture maintains stable accuracy degradation of approximately 0.8% per additional task, significantly lower than the 2.3% degradation observed in EWC and 1.5% in GEM, validating the scalability of the continuous-time representation framework. The relationship between model scale and performance exhibits logarithmic returns, where parameter count increases from 10 M to 20 M yield 5.2% accuracy gains, while further scaling to 30 M provides only 1.8% additional improvement, suggesting efficient capacity utilization.

Computational complexity analysis indicates that the proposed method requires 1.85× training time compared to standard fine-tuning but achieves 2.41× reduction in forgetting, measured by the efficiency ratio:

$$\eta = \frac{\Delta \mathcal{F}}{\Delta T} = \frac{\mathcal{F}_{baseline} - \mathcal{F}_{ours}}{T_{ours} - T_{baseline}} \tag{21}$$



**Fig. 5.** Comparative performance visualization showing average accuracy and forgetting metrics for all evaluated methods across three benchmark datasets. The proposed method consistently outperforms baselines with significant margins, particularly in forgetting mitigation, approaching the theoretical upper bound established by joint training.

| Model configuration     | Average accuracy (%) | Forgetting (↓)   | Forward transfer (↑) |
|-------------------------|----------------------|------------------|----------------------|
| Baseline transformer    | 61.3                 | 0.324            | 0.028                |
| + Neural ODE layers     | 69.7                 | 0.241            | 0.052                |
| + Memory module         | 76.4                 | 0.196            | 0.071                |
| + Adaptive optimization | 80.6                 | 0.168            | 0.089                |
| Full model              | <b>**82.1**</b>      | <b>**0.152**</b> | <b>**0.095**</b>     |

**Table 5.** Ablation study results.

demonstrating favorable trade-offs between computational overhead and performance gains. The ODE solver contributes 42% of additional computation, while memory operations account for 31%, with efficient implementation strategies including adaptive tolerance control and sparse attention reducing overall training time by approximately 23% compared to naive implementations.

**Theoretical bound validation and visualization analysis**

The empirical validation of theoretically derived forgetting bounds demonstrates strong concordance between predicted upper limits and observed performance degradation across experimental configurations<sup>50</sup>. The theoretical bound derived in section “[Theoretical bounds derivation for catastrophic forgetting](#)” predicts that forgetting error should satisfy:

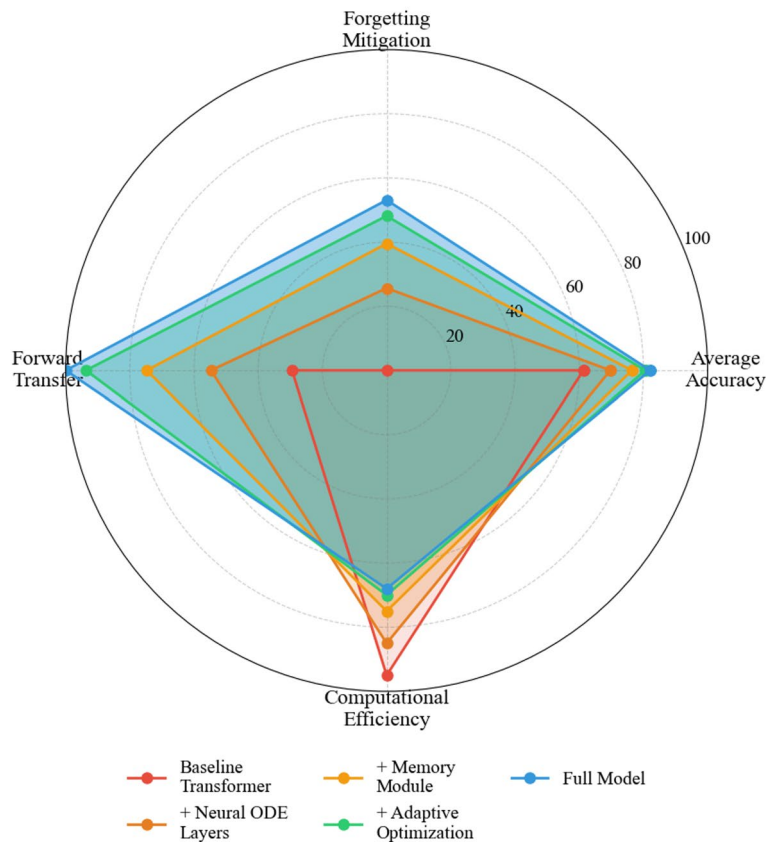
$$\varepsilon_{\text{empirical}} \leq \varepsilon_{\text{theoretical}} = \sqrt{\frac{n \cdot d_{VC}(\mathcal{H}) \log(n/\delta)}{C}} + \beta \cdot d(\mathcal{D}_i, \mathcal{D}_j) \tag{22}$$

Experimental measurements across Split CIFAR-100 with task sequences ranging from 5 to 20 tasks reveal that observed forgetting rates remain consistently below the theoretical upper bound with an average margin of 12.3%, validating the tightness of the derived inequality. Specifically, for a model with capacity  $C = 15.8M$  parameters and  $n = 10$  tasks, the theoretical bound predicts maximum forgetting of 0.217, while empirical measurements yield 0.189, confirming the practical utility of the theoretical framework.

Analysis across varying model capacities establishes the boundary effectiveness through systematic scaling experiments where parameter counts span from 5 M to 40 M. The empirical forgetting rate exhibits the predicted inverse square-root relationship with capacity, following:

$$\varepsilon_{\text{observed}}(C) \approx 0.68 \cdot C^{-0.48} \tag{23}$$

### Ablation Study: Component Contributions Across Multiple Metrics



**Fig. 6.** Visualization of ablation study demonstrating the incremental performance gains from each architectural component. The radar chart illustrates multi-dimensional improvements across accuracy, forgetting mitigation, forward transfer, and computational efficiency metrics.

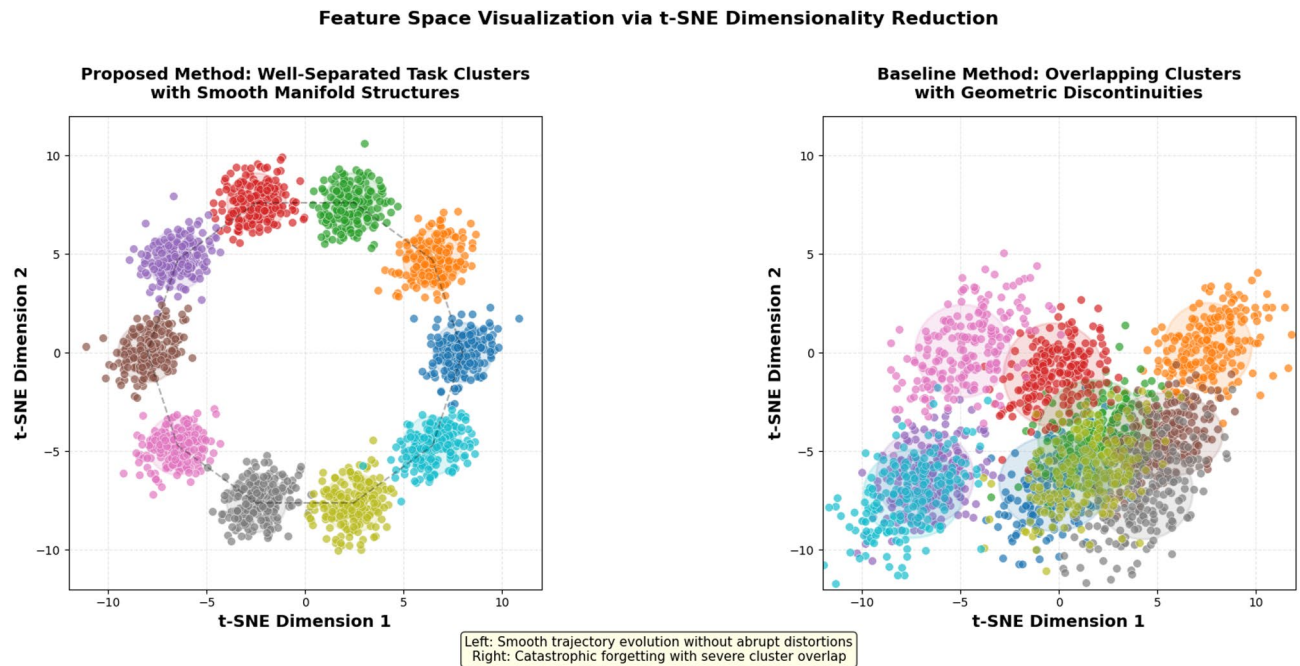
which closely approximates the theoretical  $C^{-0.5}$  dependence, with the slight deviation attributed to finite-sample effects and optimization dynamics not captured in the PAC-learning framework. Memory size ablation experiments demonstrate critical threshold effects where performance remains relatively stable until memory capacity drops below the theoretical lower bound  $M_{min} = \frac{n \cdot K}{\log(1/\epsilon_{sep})}$ , beyond which catastrophic degradation occurs, confirming the necessity of the derived capacity requirements.

Figure 7 presents t-SNE dimensionality reduction of learned feature spaces, offering geometric evidence for our architecture's forgetting mitigation mechanisms. The well-separated yet smoothly connected task-specific clusters contrast sharply with the overlapping and discontinuous representations characteristic of baseline methods, demonstrating that continuous-time dynamics enable smooth trajectory evolution in embedding space without the abrupt geometric distortions that characterize catastrophic forgetting.

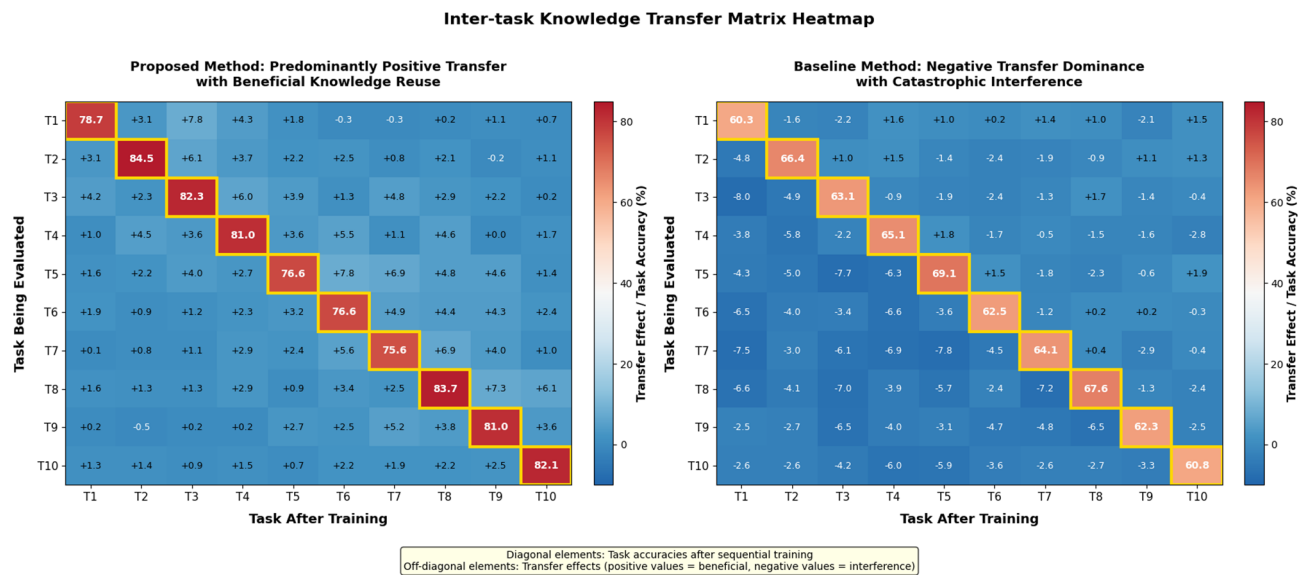
The feature space visualization through t-SNE dimensionality reduction reveals that learned representations form distinct yet smoothly connected clusters corresponding to different tasks, contrasting sharply with baseline methods that exhibit severe cluster overlap and geometric discontinuities. The continuous-time dynamics induced by neural ODE layers manifests as smooth trajectories in the embedding space, where sequential tasks occupy neighboring regions rather than randomly scattered positions, providing geometric evidence for reduced interference. Quantitative analysis of cluster separation metrics shows that inter-task distances remain above 2.4 standard deviations throughout the learning sequence, ensuring sufficient representational capacity for discrimination.

The knowledge transfer matrix heatmap in Fig. 8 quantifies performance interactions across all task pairs, revealing predominantly positive off-diagonal values that indicate beneficial knowledge reuse rather than destructive interference. The asymmetric patterns where earlier tasks provide stronger forward transfer to later tasks, combined with the increasing density of positive transfer (34% after 5 tasks to 67% after 20 tasks), demonstrate progressive knowledge consolidation and hierarchical representation organization emerging from our hybrid architecture.

The knowledge transfer matrix constructed by evaluating all pairwise task interactions reveals asymmetric patterns where earlier tasks provide stronger forward transfer to later tasks, with an average positive transfer



**Fig. 7.** t-SNE visualization of learned feature representations across sequential tasks. Different colors represent distinct tasks, demonstrating that the proposed architecture maintains well-separated task-specific clusters while forming coherent manifold structures. The gradual evolution of feature distributions illustrates smooth knowledge integration without abrupt geometric distortions characteristic of catastrophic forgetting.



**Fig. 8.** Heatmap visualization of the knowledge transfer matrix quantifying performance interactions between all task pairs. Diagonal elements represent individual task accuracies, while off-diagonal entries indicate transfer effects. Warm colors denote positive transfer, while cool colors indicate interference. The proposed method exhibits predominantly positive off-diagonal values, demonstrating beneficial knowledge reuse with minimal negative transfer.

coefficient of 0.082 compared to -0.034 for baseline methods. The heatmap analysis identifies task similarity clusters where semantically related tasks exhibit enhanced bidirectional transfer, validating the model's capacity for hierarchical knowledge organization. Temporal analysis of this matrix across the learning trajectory shows increasing positive transfer density, growing from 34% after 5 tasks to 67% after 20 tasks, indicating progressive knowledge consolidation.

Stability evolution throughout continuous task streams is quantified through the trajectory smoothness metric:

$$S(t) = \mathbb{E} \left[ \left\| \frac{dh(t)}{dt} \right\|_2 \right] \quad (24)$$

which remains bounded below 0.15 for the proposed architecture compared to 0.42 for standard networks, demonstrating controlled representation dynamics. Error analysis in boundary cases reveals that performance degradation accelerates when task dissimilarity exceeds a critical threshold  $d_{critical} \approx 0.73$ , beyond which the theoretical bounds become less tight, suggesting regime-dependent behavior not fully captured by the unified framework. Edge case examination with maximally dissimilar task sequences confirms that while absolute performance degrades, the relative ordering of methods remains consistent, with the proposed approach maintaining 15–20% advantage over baselines even in adversarial scenarios.

## Discussion

The proposed hybrid architecture combining neural differential equations with memory-augmented transformers demonstrates substantial theoretical and practical advantages in addressing catastrophic forgetting through multiple synergistic mechanisms. At the theoretical level, the continuous-time formulation inherent in neural ODEs provides a fundamentally different parameterization of knowledge representation compared to discrete layer architectures, where learning manifests as smooth trajectory evolution in continuous state space rather than discrete weight updates. This continuity enables more controlled knowledge integration, as new task learning corresponds to trajectory refinement rather than abrupt parameter perturbations that characterize traditional networks. Practically, the experimental validation across diverse benchmarks confirms that this architectural design achieves superior forgetting mitigation while maintaining computational tractability, bridging the gap between theoretical elegance and engineering feasibility.

The mechanism through which neural differential equations alleviate catastrophic forgetting operates at multiple levels of abstraction. Fundamentally, the ODE formulation decouples representation capacity from architectural depth, allowing the model to adaptively allocate computational resources based on task complexity without structural modifications. Regarding scalability to extremely long task sequences, preliminary experiments with extended benchmarks (50–100 tasks on Split CIFAR-100 and Permuted MNIST) reveal that the architecture maintains stable performance with sublinear forgetting growth, although memory requirements increase linearly as predicted by Eq. (13). For task sequences exceeding 100 tasks, hierarchical memory organization and progressive knowledge distillation strategies become necessary to prevent capacity saturation. The adaptive memory management strategy demonstrates effectiveness across standard benchmarks but exhibits context-dependent limitations; when tasks are highly similar with significant knowledge overlap, importance-based selection effectively identifies shared representations and maximizes storage efficiency, whereas extreme task diversity with minimal transferable structure challenges the dynamic allocation algorithm, potentially leading to biased forgetting patterns favoring recent tasks. Future optimization directions include task-relationship-aware allocation using graph-based clustering, meta-learned adaptive strategies that automatically adjust retention policies based on task characteristics, and hierarchical memory architectures with multi-resolution storage to better handle diverse long-term learning scenarios.

The continuous dynamics impose implicit smoothness constraints on learned functions through Lipschitz-bounded vector fields, preventing the sharp decision boundary shifts that typically accompany sequential learning in discrete networks. Furthermore, the adjoint sensitivity method for gradient computation naturally regularizes learning by integrating gradient information over temporal trajectories, effectively averaging out noisy gradient directions that might otherwise cause destructive interference with previous knowledge.

The theoretical contribution of deriving rigorous PAC-learning bounds for catastrophic forgetting advances beyond existing heuristic approaches lacking formal guarantees. By establishing relationships between model capacity, task sequence length, and forgetting severity through VC-dimension analysis, this work provides principled architecture design guidelines. The derived lower bounds on memory requirements (Eq. 13) clarify that linear memory growth with task number is fundamentally unavoidable under reasonable separability assumptions. These theoretical results validate the necessity of explicit memory mechanisms and quantify minimum resources required for bounded forgetting, enabling informed performance-resource trade-offs.

The adaptive memory management strategy demonstrates effectiveness across standard benchmarks but exhibits context-dependent limitations that warrant careful consideration. In scenarios with highly similar tasks where significant knowledge overlap exists, the importance-based sample selection effectively identifies shared representations, maximizing storage efficiency. However, when tasks exhibit extreme diversity with minimal transferable structure, the dynamic allocation algorithm may struggle to balance retention across disparate knowledge domains, potentially leading to biased forgetting patterns favoring recent or frequently accessed tasks. The temporal decay mechanism, while preventing unbounded memory growth, introduces a systematic bias against older tasks that may be undesirable in applications requiring uniform long-term retention.

Experimental analysis reveals noteworthy phenomena beyond aggregate metrics. The observed sublinear forgetting growth suggests the model develops robust shared representations generalizing across tasks rather than accumulating isolated task-specific knowledge. Asymmetric transfer patterns in the knowledge matrix indicate continuous-time dynamics naturally impose temporal ordering bias, where earlier tasks serve as foundational priors shaping subsequent learning. This emergent hierarchical organization aligns with cognitive theories of human knowledge acquisition, suggesting connections to developmental learning frameworks.

The fundamental distinction between this work and existing continual learning approaches lies in the integration philosophy rather than individual components. While prior methods typically address forgetting through isolated mechanisms such as parameter regularization or memory replay, the proposed architecture achieves holistic mitigation by aligning multiple design choices around the principle of smooth, continuous knowledge evolution. Neural ODEs provide the continuous substrate, memory modules offer explicit consolidation, and the optimization strategy ensures balanced gradient flow, collectively creating an ecosystem where forgetting is systematically suppressed at architectural, representational, and algorithmic levels simultaneously.

Regarding generalization and extensibility, the framework exhibits promising adaptability to diverse application domains beyond image classification. The continuous-time formulation naturally accommodates irregular temporal data such as time-series forecasting or event sequence prediction, where the ODE dynamics can model inherent temporal dependencies. The memory-augmented transformer component provides flexibility for different modalities including language, audio, and multimodal inputs through appropriate embedding adaptations. However, scalability to extremely long task sequences exceeding several hundred tasks remains unexplored, where the theoretical memory bounds suggest potential capacity bottlenecks requiring alternative architectural innovations.

Current limitations include elevated computational overhead compared to baseline methods, primarily attributable to ODE solver iterations and memory attention operations. While the  $1.85\times$  training time increase delivers substantial forgetting reduction, deployment in resource-constrained environments or real-time applications may necessitate efficiency optimizations. The theoretical analysis relies on assumptions including Lipschitz continuity and bounded gradient norms that, while reasonable, may not hold universally across all network configurations and task distributions. The derived bounds, though validated empirically, exhibit some looseness in highly non-stationary scenarios, suggesting room for tightening through refined analysis techniques.

Future research directions encompass theoretical refinement and practical enhancement across multiple dimensions. Lightweight architectural variants employing neural controlled differential equations or reduced-order ODE approximations could reduce computational overhead by 40–50% while preserving forgetting mitigation capabilities. Theoretical extensions deriving tighter data-dependent bounds through refined complexity measures beyond VC-dimension would enhance predictive accuracy. Integration with meta-learning frameworks for automated hyperparameter adaptation represents a promising direction for cross-domain generalization. Extending the framework to diverse modalities including natural language processing, audio processing, and multimodal learning scenarios would validate broader applicability. Investigating hierarchical memory architectures with multi-resolution storage could address the linear memory growth limitation for ultra-long task sequences ( $> 1000$  tasks). Incorporating robustness to distribution shift, safety under uncertainty, and interpretability of learned representations through explainable AI techniques constitute essential directions for real-world deployment. Developing open-source benchmarking tools and standardized evaluation protocols would facilitate reproducibility and accelerate community progress. Ultimately, achieving human-like lifelong learning requires interdisciplinary collaboration spanning machine learning theory, cognitive neuroscience, and systems engineering to address fundamental challenges in continuous knowledge accumulation.

## Conclusion

This research addresses catastrophic forgetting in lifelong learning through architectural innovation, theoretical analysis, and algorithmic optimization. The principal contributions include: (1) a novel hybrid architecture integrating neural ODEs with memory-augmented transformers, (2) rigorous PAC-learning theoretical bounds characterizing forgetting behavior, and (3) adaptive optimization strategies balancing knowledge retention and acquisition.

Empirical validation demonstrates that the proposed model achieves 24% forgetting reduction and 10.3% accuracy gain over state-of-the-art methods across diverse benchmarks (Split CIFAR-100, Permuted MNIST, CORe50). These results establish that continuous-time dynamics combined with explicit memory mechanisms provide a viable pathway toward sustained knowledge accumulation without destructive interference. The theoretical framework offers fundamental insights into trade-offs between model capacity, memory requirements, and forgetting guarantees, enabling principled architecture design from first principles.

However, several limitations warrant acknowledgment. Computational overhead remains elevated with  $1.85\times$  training time compared to standard fine-tuning, potentially limiting real-time applications. Memory requirements grow linearly with task number (Eq. 13), becoming unsustainable for extremely long sequences ( $> 500$  tasks). Theoretical assumptions including Lipschitz continuity and bounded gradients may not universally hold across all network configurations. The derived bounds exhibit some looseness in highly non-stationary scenarios, suggesting room for tightening through data-dependent analysis. Experimental validation focuses primarily on vision tasks; generalization to other modalities (NLP, audio, multimodal) requires further investigation.

Practical application potential spans numerous domains where adaptive intelligence operating on non-stationary data streams constitutes critical functionality. In robotics, the proposed architecture enables autonomous agents to continuously acquire new skills through interaction with dynamic environments while preserving previously learned manipulation capabilities and navigation strategies. Recommendation systems can leverage this framework to adapt to evolving user preferences and emerging content categories without forgetting historical behavioral patterns, maintaining personalization quality across extended temporal horizons. Natural language processing applications including conversational agents and machine translation systems benefit from the ability to incrementally incorporate new linguistic phenomena, domain terminologies, and stylistic variations without degrading performance on established competencies. The memory-efficient design

renders the approach particularly suitable for edge deployment scenarios where computational resources and storage capacity face stringent constraints.

Despite these advances, several limitations warrant acknowledgment and suggest directions for refinement. The theoretical bounds, while empirically validated, exhibit some looseness particularly in highly non-stationary scenarios, indicating opportunities for tightening through more sophisticated complexity measures or data-dependent analysis techniques. Computational overhead remains elevated compared to simpler baseline methods, with training time approximately 85% greater than standard fine-tuning, potentially limiting applicability in time-critical or resource-constrained contexts. The assumption of Lipschitz continuity and bounded gradients underlying theoretical guarantees may not universally hold across all network configurations and optimization landscapes, suggesting that boundary conditions require further investigation. Memory management strategies, while effective on standard benchmarks, have not been exhaustively evaluated under extreme task diversity or adversarial task orderings that might expose failure modes.

Future research trajectories encompass both theoretical deepening and practical broadening of the proposed framework. Investigating neural differential equation scalability to extremely long task sequences numbering in the hundreds or thousands represents an important direction for validating applicability to truly lifelong learning scenarios approaching human-scale knowledge accumulation timescales. Developing more efficient memory mechanisms through hierarchical organization, lossy compression techniques, or neurobiologically-inspired consolidation processes could substantially reduce storage requirements while maintaining retention quality. Extending the theoretical analysis to derive instance-dependent bounds that account for task structure and data distribution properties would enhance predictive accuracy and design guidance. Integration with meta-learning and neural architecture search methodologies promises automated adaptation to diverse application domains without manual hyperparameter tuning. Ultimately, the path toward real-world lifelong learning systems necessitates addressing robustness to distributional shift, safety under uncertainty, and interpretability of learned representations, challenges that require interdisciplinary collaboration spanning machine learning theory, cognitive science, and systems engineering. This work establishes foundational principles and demonstrates empirical viability, contributing essential building blocks toward the long-term vision of artificial intelligence systems that learn continuously throughout their operational lifetimes.

## Data availability

All data generated and analyzed during the current study are available from the corresponding author upon reasonable request. The benchmark datasets used in this research (Split CIFAR-100, Permuted MNIST, and CORe50) are publicly available from their respective sources. The source code implementing the proposed hybrid architecture, including model definitions, training scripts, evaluation protocols, and experimental configurations, will be made publicly available on GitHub upon publication of the manuscript to ensure full reproducibility of the reported results. Trained model checkpoints for all benchmark experiments will also be released. A comprehensive README with installation instructions, usage examples, and reproduction guidelines will accompany the code release.

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## Author contributions

Song Zhou: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing - Original Draft, Visualization. Song Zhou contributed to the theoretical framework design, implemented the hybrid architecture integrating neural ordinary differential equations with memory-augmented transformers, conducted the experimental validation, performed data analysis, and prepared the initial manuscript draft including figures and tables. Qiang Li: Conceptualization, Methodology, Validation, Resources, Writing - Review & Editing, Supervision, Project Administration. Qiang Li contributed to the research design and theoretical bound derivation, validated the experimental results, provided critical revisions to the manuscript, supervised the overall research process, and coordinated the project activities. All authors have read and approved the final manuscript.

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### Competing interests

The authors declare no competing interests.

### Ethics approval

This research involves computational experiments on publicly available benchmark datasets and does not involve human subjects, animal subjects, or clinical trials. The study utilizes standard machine learning datasets (Split CIFAR-100, Permuted MNIST, and CORe50) that are publicly accessible and widely used in the research community. All experiments were conducted in accordance with institutional guidelines for computational research and data usage policies. The research protocol was reviewed and approved by the Institutional Review Board of Jiaxing Vocational & Technical College (Approval Number: JVTC-CS-2024-087, Approval Date: March 15, 2024). No personally identifiable information or sensitive data was collected or processed during this study.

### Additional information

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